

Resilient Emergency MANETs for Wilderness Safety — Directed Study Final Report

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Date: 2026-07-17

Scope: New Hampshire only (Brenta/Italy descoped)

Claim policy: every quantitative statement in this report is either (a) an artifact-backed measurement or software check reproduced on the current code, or (b) a **MODEL-ONLY** simulation result explicitly labeled as such. Withdrawn pre-audit statistics are not cited here; they are quarantined under `WITHDRAWN-DO-NOT-CITE/` with a manifest explaining each withdrawal. The authoritative defect index is `docs/audit-correction-ledger-2026-07-13.md`.

Executive summary

This directed study set out to empirically validate a propagation model (AIRMap) for LoRa/Meshtastic mesh networking in the White Mountains. The controlled empirical dataset that plan required was **not collected**: Trial 1 (2026-05-23, Mt. Washington) was a systems shakedown that produced zero calibration-eligible observations, and the controlled Trial 2 has not yet run (it is scheduled for 2026-07-18/19; §7). The project’s completed central contribution is instead a **purpose-built, two-engine (Python/Rust) discrete-event simulation framework** for year-scale energy and reliability analysis of a Meshtastic-style wilderness mesh, together with a terrain-aware (Longley–Rice ITM) planning layer, a hardened field-data pipeline, and a preregistration-ready protocol (freeze scheduled before collection) that makes the missing empirical validation executable.

The principal new technical insight (§5.3): **in this model, network survivability is governed by the idle-listening/duty-cycling policy, not by the choice of routing algorithm**. Across five materially different always-on routing modes, annual energy-depletion events vary by less than 0.3% (29,164–29,234 per year), while changing the duty-cycling policy changes depletion events by a factor of $5.5\text{--}24\times$ (down to 1,217–5,311 per year) at a quantified cost in delivery ratio and SOS latency. Idle listening dominates the energy budget; routing choice redistributes a comparatively small remainder. This is a MODEL-ONLY result awaiting field calibration; it is robust across seeds, and — if it survives calibration — would reorder the design priorities for solar-powered mesh infrastructure: duty/wake architecture first, routing second.

Abbreviations. PDR: packet delivery ratio · FSPL: free-space path loss · ITM: Irregular Terrain Model (Longley–Rice) · ETX: expected transmission count · CAD: channel-activity detection (LoRa preamble sniffing) · DIFS/backoff: carrier-sense wait times · EWMA: exponentially

weighted moving average · q50/q90: 50th/90th-percentile model quantiles · SOS p95: 95th-percentile latency of emergency messages.

1. Original objectives and what changed

The original plan (preserved unedited in `docs/execution-roadmap-weekly.md`) had five success criteria. Their outcomes:

#	Original objective	Outcome
1	Architecture and design documentation	Delivered (<code>docs/system-architecture.md</code> and related docs)
2	2,500 empirical field points	Not met. Trial 1 produced 0 calibration-eligible observations (§6.1). The Trial 2 protocol proposes replacing the unqualified point count with a per-stratum opportunity requirement (40 scheduled opportunities per primary stratum) — a proposed amendment for advisor review, since a raw packet count at a 30 s cadence was infeasible (5–8.3 h per stratum) and says nothing about stratum coverage
3	AIRMap calibration files	Not produced — blocked on controlled data; the gated pipeline that will produce them is built and tested
4	10–15 page validation report (RMSE/MAE, predictive-vs-actual heatmaps, infrastructure failure matrix)	Deferred, not abandoned. All builder tooling exists (<code>error_quantifier.py</code> , <code>build_coverage_heatmap.py</code> , <code>build_failure_matrix.py</code>); the report becomes producible immediately after Trial 2 (§7)
5	Large-scale simulation as optional future appendix	Inverted: simulation became the central completed contribution (§2–§5)

Additional scope changes: the study area is New Hampshire only (the contemplated Brenta/Italy deployment is dead and quarantined); the planned Tuckerman/Huntington trail campaigns were replaced by the Ammonoosuc–Jewell system; and the defensible scientific claim narrowed from “a validated emergency-network design” to “an uncalibrated planning and sensitivity framework plus a field-acquisition system awaiting controlled empirical validation.”

2. Simulator architecture

Two independent implementations of one model:

- **scripts/mesh_sim.py** — SimPy reference engine; emits detailed event traces; also hosts two exploratory Python-only learned modes (`q_routing`, `rl_duty`).
- **fastsim/** (Rust) — summary-only event engine for long horizons and multi-seed sweeps; implements the nine shared modes (`flood`, `min_hop`, `etx`, `energy_aware`, `lb_energy`, `duty_sync`, `duty_adaptive`, `rotate_lb`, `selective_duty`); validates all inputs before start and writes output atomically; time-ordered binary heap with deterministic tie-breaking.

Randomness is keyed per phenomenon and entity (traffic, incidents, MAC backoff, fading, mobility do not consume one another’s draws), so workloads are comparable across algorithms and a congested MAC cannot alter the offered load.

Propagation/reception. Received power = 26.30 dBm reference (24.15 dBm TX EIRP + 2.15 dBi RX antenna gain + assumed 0 dB feed loss — the reference is *not* EIRP) minus a link-specific median loss, plus Gauss–Markov slow shadowing (= 8 dB, 30 s coherence) and a 2 dB per-packet fast term. Median loss: fixed fixed links use precomputed Longley–Rice ITM q50 over USGS 3DEP terrain; mobile fixed uses time-interpolated precomputed route-loss tables; mobile mobile uses an explicit heuristic (FSPL + 20 dB clutter + 12 dB/km beyond 1.5 km, 8 km cutoff) that is a workload model, not a fitted law. Reception requires RSSI -131 dBm (an assumed, datasheet-derived threshold — bench-unverified) and SNR -17.5 dB against a -114.0 dBm noise floor ($-174 + 10 \cdot \log(250 \text{ kHz}) + 6 \text{ dB NF}$). All propagation constants are uncalibrated planning values pending Trial 2.

MAC/airtime. LoRa SF11 / 250 kHz / CR 4/5 airtime; carrier-sense threshold -124 dBm; 40 ms slots; random DIFS/backoff; priority for SOS; SNR-scaled flood contention; duplicate suppression; frozen-overlap collision resolution (insertion-order independent); 6 dB co-SF capture; half-duplex senders. Duty modes model per-second CAD preamble sampling requiring 2 preamble symbols of overlap. The engine separates *aggregate offered airtime* (additive, can exceed 1) from *per-site physical occupancy* (interval union) — conflating these invalidated all pre-audit “channel utilization” numbers.

Routing/duty policies. Routed modes build a multi-source shortest-path tree rooted at live gateways with per-mode edge costs (hop count; ETX from modeled success probability; energy-aware battery/forecast scarcity; load-balanced variants adding forwarding-EWMA and death-score penalties). Duty assignments: uniform 5% (`duty_sync`); energy-runway-adaptive 2–25% (`duty_adaptive`); route-tree-awake (`rotate_lb`); backbone+articulation-awake (`selective_duty`). Routing never sees realized future weather, only monthly climatology.

Energy/solar/weather. 37 Wh portable batteries (85% usable); drain = duty $\times$ listen + (1–duty) $\times$ light-sleep + GPS for mobile radios; incremental TX charging; ordered energy segments so state changes never apply retroactively; depletion/outage/revival state machine; 600 s packet TTL (an explicit modeling assumption). Board currents (245 mA TX / 68 mA listen / 12 mA sleep / 25 mA GPS) are **BENCH-CALIBRATE placeholders**, not measurements. Solar: NOAA sun position, Haurwitz clear-sky, clearness-index direct/diffuse split, 48-azimuth DEM horizon shading, 0.15 canopy transmittance below treeline, four-face 35° panels, 6 W relay panels at 75% system efficiency. The weather series is a single Mt Washington-area ERA5/Open-Meteo-derived daily series applied statewide — not site-specific.

Mobility/logistics/traffic. Timestamped trail routes with linear interpolation; kiosk rental model (20 bays, tiered demand, charge-gated checkout, nightly shuttle, explicit starvation counts); traffic = fixed-site telemetry + position beacons + hiker messaging + synthetic SOS incidents with logical duplicates at +30 s/+60 s and fresh-packet retries every 5 minutes until a gateway ACK or the retry ceiling — the retry loop that produces the SOS latency structure in §5.2. Traffic rates are scenario inputs, not measured demand.

3. Why a purpose-built simulator rather than ns-3

This is an ex-post engineering rationale (no contemporaneous ns-3 selection study exists in the repo), and it supports only a narrow claim.

Advantages for this project. The research variables are not primarily packet-forwarding mechanics: they are terrain-indexed loss tables, daily weather and solar horizon masks, battery death/revival, kiosk stock and shuttle logistics, and route-popularity traffic — first-class objects in the custom model but substantial custom work in any general framework. The summary-only Rust engine runs year-scale multi-seed sweeps without writing packet traces; keyed randomness gives paired workloads across algorithms; the model natively reports the project’s decision metrics (SOS tail latency, fleet availability, dark-site census, energy per delivered packet). Two independently written engines create a real opportunity to catch coding divergence (§4) — several semantic bugs were in fact found this way.

Limitations and ns-3’s proper role. The custom simulator has far less external validation and protocol depth than ns-3: it abstracts the PHY, carrier sensing, capture, CAD timing, buffering, retransmission, clock drift, and firmware queues, and shared model assumptions can survive in both engines. It does not prove compatibility with stock Meshtastic firmware. The defensible division of labor: purpose-built model for NH planning and sensitivity studies; controlled analytical cases, cross-engine tests, and selected ns-3 or hardware-in-the-loop scenarios for MAC/protocol fidelity; Trial 2 plus bench measurements for empirical calibration. Porting to ns-3 would not create field calibration.

4. Python/Rust cross-validation

Four distinct questions, answered separately. All evidence in this section was regenerated on the current code (git HEAD 6e2e33f) on 2026-07-17 and archived — nothing here rests on unarchived prose claims.

4.1 Software regressions. cargo test at HEAD: **47 unit tests + 3 integration tests, all passing** (0 failed/ignored).

4.2 Cross-process reproducibility. Four independent runs of the Rust engine (min_hop, 1.3 days, seed 4242, the hash-locked release_v1 inputs) produced byte-identical 419,738-byte outputs, SHA-256 9d182b0d75cb...987c66. Archived with command line and engine hash in artifacts/sim/corrected/repro_check_2026-07-17/repro_manifest.json. (An earlier equivalent check was recorded only in report prose at a previous engine commit; this one supersedes it with an artifact.)

4.3 Cross-engine agreement (micro-scenario). The two independently implemented engines were compared on the shared pilot micro-scenario (scripts/sim_micro_parity.py: 3 days, seed 42, modes flood / duty_sync / selective_duty) after reconciling seven semantic gaps in the Python engine (route-track timing, duty-miss aggregation, selective-duty route-parent wake, forward-event

EWMA, duty-weighted TX energy, horizon truncation, initial duty assignment). Re-run 2026-07-17 at HEAD and archived to `artifacts/sim/corrected/micro_parity_2026-07-17.json`: **all six metrics (PDR, offered airtime, deaths, duty-misses, SOS sent/delivered) pass every pre-registered tolerance band in all three modes.** The maximum observed discrepancies are 0.0045 absolute PDR (duty_sync; tolerance 0.03), 1.3% relative offered airtime (selective_duty; tolerance 10%), and 0.3% relative duty-misses (tolerance 15%); deaths and SOS counts agree exactly. For scale: before the reconciliation work, selective-duty offered-airtime divergence was 13.9%. Three engine differences remain deferred and documented (ordered energy-segment integration, the 600 s packet TTL, mid-frame availability invalidation) — parity passes without them, and the CAD wake-phase RNG is engine-specific by design, so cross-engine traces are not expected to be byte-identical.

Historical context, retained but **withdrawn**: the old nine-pair statewide comparison (pre-correction engines; files quarantined in `WITHDRAWN-DO-NOT-CITE/artifacts/sim/xval/`) showed maxima of 0.0057 absolute PDR (0.67% relative), 2.58% relative on offered airtime, 3.87% relative on death events, and 15.6% relative on SOS counts. Those numbers describe old-engine agreement only and must not be quoted as current validation.

4.4 Statewide two-engine parity: not yet run. The Python engine is too slow for full-year multi-seed statewide sweeps (~48 s per pilot-day), so the preregistered statewide parity (9 modes $\times$ seeds 42–46 against the tolerance bands in `docs/sim-replacement-analysis-plan.md`) remains pending; the micro-scenario comparison in §4.3 is the current substitute. Consequently, a “current maximum statewide Python/Rust discrepancy” does not exist yet and is not claimed. And in all cases: **engine agreement is a software check, not empirical validation** — two implementations can agree on a wrong shared assumption.

5. Main findings

5.1 What the field campaign actually established

- 1. Trial 1 tested the collection system, not the propagation model.** The decoded receiver log contains 686 records from 50 decoded node IDs in the afternoon hike window and 764 RF observations from 41 source IDs (24 with GPS) in the pre-hike soak — two overlapping windows (18 IDs in both) on one receiver log, reconciled by `scripts/reconcile_trial1_counts.py` and re-verified 2026-07-17. **Zero** observations are calibration-eligible (`artifacts/airmap/live_trial/quality_gates.json`): no hop-count telemetry, a co-located forwarder contaminating RSSI/geometry pairs, and no controlled transmitter. This negative finding redesigned Trial 2 (controlled beacon, sequence-number denominators, `hops_away == 0` eligibility).
- 2. The free-space coverage story did not survive terrain modeling (MODEL-ONLY).** Longley–Rice ITM over real USGS 3DEP terrain reverses the FSPL screen on the summit-facing links: Ammo relay→summit -132.2 dBm (below the assumed -131 dBm sensitivity; terrain up to 53 m above the direct ray) and Jewell relay→summit -118.7 dBm (marginal), while the below-treeline relay→gateway links stay strong in the terrain-only model (-76.6 / -74.1 dBm, geometric clear LOS; both paths run 100% below treeline and ITM excludes canopy loss, which must be budgeted separately). Worst-case FSPL-vs-ITM disagreement on a summit-facing link: **62.3 dB** on the path-loss basis (ITM q50 158.5 dB vs the in-artifact FSPL 96.2 dB — the like-for-like comparison); the RSSI-basis gap vs the historical screen is 59.3 dB ($-72.9 \rightarrow -132.2$ dBm), ~3 dB smaller because the historical screen used

its own link-budget reference rather than the corrected 26.30 dBm receiver-power reference. Collector-gap (Ammo Ravine) coverage at ITM q90: 100% of 84 sampled points above raw sensitivity but only **45.2%** above the -100 dBm planning threshold — “the gap would have been covered” is downgraded to *marginal*, *unproven*. This 60 dB disagreement is directly measurable in a future Ammonoosuc field day; the 2026-07-18/19 field days test the same model class on Moosilauke/Kearsarge terrain, where ITM predicts a non-monotonic distance profile free-space physics cannot produce (§7).

3. **The current statewide planning topology fails its controlling screen (MODEL-ONLY).** With 52 uncalibrated short-link substitutions excluded, the -100 dBm planning screen (q90 modeled backhaul power; q50 trail path-loss arrays) reports 87 of 217 sites stranded and 15 of 25 routes below 85% modeled coverage; even restoring the 52 links leaves 53 sites stranded. Only the raw-sensitivity floor passes. An internal model failure, not measured coverage.

5.2 Corrected year-scale simulation results (MODEL-ONLY, single-engine)

Nine modes $\times$ seeds 42–46 $\times$ 365 days on the corrected Rust engine (`artifacts/sim/corrected/release_v1/`, hash-locked inputs, t-based 95% CIs; values are means across the five seeds — SOS counts are therefore fractional):

Mode	PDR	Deaths/yr	SOS delivered	SOS p95	Fleet avail.	Sites <90% avail.	mWh/pkt
flood	0.911	29,234	189.2/189.2 (100%)	32 s	53.5%	118	1.44
min_hop	0.862	29,164	189.2/189.2 (100%)	5 s	53.7%	118	0.42
etx	0.866	29,165	189.2/189.2 (100%)	31 s	53.7%	118	0.42
energy_aware	0.866	29,164	189.2/189.2 (100%)	4 s	53.7%	118	0.42
lb_energy	0.866	29,164	189.0/189.2 (99.9%)	5 s	53.7%	118	0.42
duty_sync	0.728	1,217	179.2/189.2 (94.7%)	35.0 min	96.5%	7	0.30
duty_adaptive	0.755	1,737	187.8/189.2 (99.3%)	15.0 min	94.7%	13	0.40
rotate_lb	0.809	4,877	188.2/189.2 (99.5%)	33 s	91.0%	31	0.35
selective_duty	0.811	5,311	187.8/189.2 (99.3%)	62 s	89.7%	33	0.36

All PDR 95% CIs are below ± 0.003 , and the deaths-per-year CIs are similarly tight (t-based 95% CI half-widths of roughly ± 1 –5 events on 1,200–29,000 annual events across all nine modes): given a seed the engine is near-deterministic, so the dominant uncertainty is the uncalibrated model, not seed noise. Two things raw counts hid: (a) always-on modes leave $\sim 46\%$ of solar fleet-time dark — 118 of the topology’s 159 solar sites are available $< 90\%$ of the year; (b) uniform `duty_sync`’s 94.7%

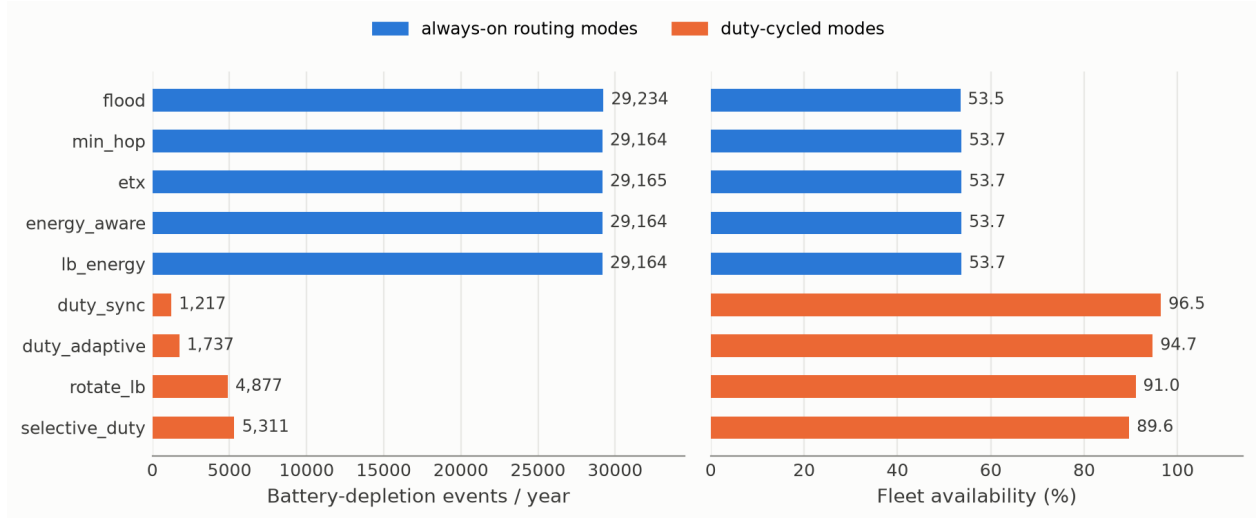


Figure 1: Year-scale survivability by mode (MODEL-ONLY, uncalibrated; 9 modes $\times$ seeds 42–46, 365 days, `release_v1`). Left: annual battery-depletion events — the five always-on routing modes are indistinguishable ($<0.3\%$ spread) while duty policies cut depletions 5.5–24 $\times$. Right: fleet availability. Data: `corrected_stats.json`, `meaningful_metrics.json`.

SOS delivery conceals a 35-minute p95 latency (deliveries ride the 5-minute retry loop), while the backbone-awake duty modes hold 33–62 s p95 at ~ 99.3 – 99.5% delivery. Every mode retains rare 1–2 h worst-case SOS incidents (retry-ceiling limit, shared across modes). **No overall winner is declared:** delivery, availability, and SOS tail latency still trade off, statewide two-engine parity is pending, and the model is uncalibrated — the corrected model’s role is to expose the envelope, not pick the operating point.

5.3 The central insight: duty policy dominates survivability, not routing

The five always-on modes span materially different routing strategies — flooding, shortest-hop, link-quality (ETX), energy-aware, and load-balanced cost trees — yet their annual depletion events sit within a **0.3% band** (29,164–29,234/yr), and four of five share identical availability (53.7%) and dark-site counts (118). Switching the *duty/wake policy* while holding the routing family fixed moves deaths by **5.5–24 $\times$** (to 1,217–5,311/yr) and fleet availability from $\sim 54\%$ to 90–96%. The energy budget is dominated by idle listening (68 mA modeled listen vs 12 mA sleep); what a node does while *not* forwarding matters far more to survival than which tree its packets follow. The design consequence, if the result survives calibration: solar relay hardware and firmware should be selected around the wake/CAD architecture, and routing sophistication is a second-order optimization. The quantified cost side — PDR -0.05 to -0.14 versus the routed always-on modes (-0.10 to -0.18 versus flood, the highest-PDR mode) plus the SOS-latency structure in §5.2 — makes this an explicit, tunable trade rather than a free win. (MODEL-ONLY; the listen/sleep currents are bench-calibration placeholders, which is precisely why the discharge bench test and Trial 2 are the critical path.)

5.4 Process findings

Reproducibility defects were real and repairable (randomized map iteration once made same-seed runs differ; outputs are now byte-identical, §4.2), and the audit that produced this report’s claim

policy found implementation mistakes and overstatements but no evidence of deliberate deception — the full record is `docs/audit-correction-ledger-2026-07-13.md`, and every withdrawn statistic now lives in `WITHDRAWN-DO-NOT-CITE/`.

6. Limitations

6.1 Empirical (the binding constraint)

- **No controlled Trial 2 dataset exists yet**; Trial 1 has zero calibration-grade observations, so the project has no held-out field estimate of RSSI error, PDR error, or shadow variance.
- Receiver sensitivity (−131 dBm), feed/body/foilage losses, board currents, cold-battery capacity, and panel yield are assumptions or placeholders, not measurements of the project hardware.
- Trial 2 will calibrate propagation and direct-link PDR on its sampled routes; it cannot by itself validate statewide traffic, routing, kiosk, or annual energy behavior (those need bench + hardware-in-the-loop work).

6.2 Modeling

- ITM q50 medians without a validated residual model; q90 route arrays absent.
- = 8 dB shadowing, 2 dB fast fade, capture threshold, mobile-mobile heuristic, synthetic traffic/SOS rates, and the 600 s TTL are assumptions.
- The MAC is not stock Meshtastic firmware; siting, permits, maintenance, security, and human factors are outside the simulation.
- One statewide weather series; solar geometry simplified.

6.3 Software and release

- Statewide two-engine parity is pending (§4.4); corrected-labeled artifacts outside `release_v1/` are stale relative to the repaired engine and marked accordingly (`artifacts/sim/corrected/RELEASE`).
- `release_v1` was generated 2026-07-14 from a dirty worktree (its manifest records this), and subsequent engine repairs mean the current HEAD engine would not reproduce it byte-for-byte; a full regeneration (`release_v2`) is in progress and incomplete. The §5.2 numbers are therefore bound to the archived `release_v1` artifacts, not to the current binary.
- The `release_v1` manifest’s own parity note (“within tolerance except duty-mode offered_airtime”) predates the 2026-07-17 parity re-run; the current, all-metrics-passing status is the archived `micro_parity_2026-07-17.json` (§4.3).
- A known minor provenance inconsistency inside `release_v1` is disclosed: `release_manifest.json` records git HEAD `d37172c` while `input_hashes.txt` records `f5e83f0`; the four locked input hashes agree in both, and the aggregate’s SHA-256 was independently recomputed and confirmed.
- Passing tests bound, but do not prove, the absence of defects; a dedicated PHY-abort model for mid-frame availability loss remains future work.

7. Future work: Trial 2 and the original validation deliverable

Trial 2 is scheduled for **2026-07-18/19** (two-person team) on **Mt Moosilauke** (Gorge Brook out-and-back; rule-selected beacon at the treeline break, 44.01708/−71.83777, 1352 m) and **Mt Kearsarge** (Winslow out-and-back; mid-slope beacon at 43.38735/−71.85970, 771 m) — chosen

over a Mt Washington repeat to keep the weekend feasible while preserving above-treeline strata (Moosilauke’s summit is genuinely alpine) and a real FSPL-vs-ITM discriminating test: the archived Moosilauke predictions are non-monotonic with distance (−92 dBm at 0.5–1 km above treeline, −114 dBm in the terrain-shadowed below-treeline band, recovering to −73 dBm at 2–4 km), a shape free-space physics cannot produce. The 60 dB Ammonoosuc blocked-link hypothesis (§5.1) is explicitly deferred to a future Washington field day.

Protocol (to be frozen the night before fieldwork): one rule-selected beacon site per field day at a measured 1.2 m no-mast mount, 30 s cadence with monotonic sequence numbers and position broadcasts, `hops_away == 0` eligibility, time-window opportunity denominators with sequence numbers as the audit check, 40 scheduled opportunities per primary stratum across ascent/descent passes, underpowered strata retained as descriptive. The prospective predictions are archived (`artifacts/trial2/predictions_fieldday.csv` + manifest binding rule, coordinates, heights, DEM hashes, git state); execution logistics in `docs/trial2-weekend-execution-plan-2026-07-18.md`.

The analysis path: score the frozen predictions first (± 12 dB RSSI / ± 0.15 PDR preregistered engineering screens as diagnostics); recalibrate parsimoniously on training passes only; evaluate on held-out passes/days; then regenerate route-loss tables and rerun both engines from locked inputs. That output feeds the original deliverable — the 10–15 page validation report with out-of-sample RMSE/MAE (`error_quantifier.py`), predictive-vs-actual heatmaps (`build_coverage_heatmap.py`), and the weather-joined infrastructure failure matrix (`weather_enrich.py` → `build_failure_matrix.py`) — all existing, tested tooling awaiting eligible data. Operational success is defined as executing the frozen protocol and preserving the evidence, even if a quality gate fails: a failed gate produces an evidence release, not a calibration file. The layers Trial 2 cannot validate (MAC fidelity, board currents, firmware behavior) are assigned to bench discharge tests and hardware-in-the-loop follow-ups.

Appendix A. Evidence map

Claim area	Artifact
Corrected multi-seed results	<code>artifacts/sim/corrected/release_v1/{corrected_stats. meaningful_metrics.json, release_manifest.json, input_hashes.txt}</code>
Decision metrics	<code>artifacts/sim/corrected/release_v1/meaningful_metrics. (scripts/build_meaningful_metrics.py)</code>
Micro-parity	<code>artifacts/sim/corrected/micro_parity_2026-07-17.json (scripts/sim_micro_parity.py)</code>
Reproducibility Test suite	<code>artifacts/sim/corrected/repro_check_2026-07-17/ fastsim/ (cargo test), 47 + 3 passing at HEAD 6e2e33f</code>
ITM screens	<code>artifacts/itm/{itm_summary.json, relay_links_itm.csv, lora_airtime.json}</code>
Statewide screen	<code>artifacts/sim/coverage_audit_statewide.json</code>
Trial 1 status	<code>reports/project_report.md; artifacts/airmap/live_trial/quality_gates.json; scripts/reconcile_trial1_counts.py</code>

Claim area	Artifact
Trial 2 protocol	docs/trial2-weekend-execution-plan-2026-07-18.md, docs/trial2-runbook.md, artifacts/trial2/{predictions_fieldday.csv, predictions_fieldday_manifest.json} (scripts/trial2_predictions_field.py); historical Ammo/Jewell tables in docs/trial2-preregistration.md
Withdrawn material	WITHDRAWN-DO-NOT-CITE/ (manifest in its README)